Agentic Fraud Detection Framework- Multi Agent Fraud Detection

Global IME Hackathon

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Abstract- Nepal is currently witnessing a massive shift toward digital payments, yet the infrastructure for real-time fraud detection remains largely reactive. This paper presents an agentic multi-model framework designed specifically for the Nepali fintech context, where mobile wallet and QR transactions have reached unprecedented volumes. Our system routes each transaction through four specialized agents—velocity, geolocation, behavioral, and graph—coordinated by a context-aware orchestrator. Unlike traditional static systems, our synthesis agent utilizes dynamic weighting that adapts based on the transaction type (e.g., P2P remittance vs. merchant QR). By combining high-performance classification with unsupervised anomaly detection, the framework achieves a real-time "approve/OTP/block" verdict in under 800ms. For an institution like Global IME Bank, this architecture provides a scalable defense that meets the "smart technology" requirements recently mandated by national regulators.

1. Introduction

The digital payment ecosystem in Nepal has reached a critical turning point. As of mid-February 2025, mobile banking users have grown to 26.57 million, while digital wallet users now total 25.85 million (Nepal Rastra Bank, 2081). This rapid adoption is most evident in QR-based retail payments, which processed over 26.3 million transactions in a single month (Nepal Rastra Bank, 2081). However, this growth has also expanded the surface area for cyber-enabled fraud. Recent findings suggest that fraud rings are increasingly utilizing "rapid layering"—moving illicit funds through multiple accounts and wallets within minutes to evade detection (Financial Intelligence Unit Nepal, 2024).

Most commercial banks in Nepal currently rely on rule-based systems that flag transactions based on static thresholds. These frameworks struggle to identify sophisticated behavioral anomalies, such as "test transactions" as low as Re. 1 used by fraudsters to verify active accounts (Financial Intelligence Unit Nepal, 2024). To counter this, the Nepal Rastra Bank has issued the Unified Payment System Directive 2081, which mandates the adoption of AI-driven, proactive monitoring tools (Nepal Rastra Bank, 2081). Meeting these requirements requires a system capable of "20-millisecond freshness"—identifying a threat before the transaction is finalized (Shen et al., 2023).

Technical literature suggests that supervised learning, specifically the eXtreme Gradient Boosting (XGBoost) algorithm, is highly effective for classifying known fraud patterns in tabular financial data (Chen and Guestrin, 2016). However, to detect novel "zero-day" attacks, it must be paired with unsupervised methods like Isolation Forest, which identifies anomalies by isolating them in a data forest rather than relying on labeled history (Liu et al., 2008).

Our research contributes a synergistic framework that bridges these two approaches. By implementing a dynamic ensemble strategy, our system adjusts the importance of different detection signals based on the specific transaction context (Britto, Sabourin and Oliveira, 2014). This ensures that a high-value P2P transfer is scrutinized differently than a low-value merchant payment, optimizing the balance between security and user experience for Global IME Bank's growing digital customer base.

2. Literature Review

With the increase in the digitalization of banking services and currencies, there has been notable increase in the number of digital thefts and frauds and their approach to gain financial benefits. According to the Nilson Report, global payment card fraud losses exceeded $33.83 billion worldwide in 2023 (Nilson Report, 2023). Few decades ago when there were no such technologies to digitalize the monetary transaction, fraud was still prevalent. However, the nature and scale of fraud have been transformed dramatically with the advancement of technologies and rise of Artificial Intelligence (AI). In past, fraud methods like check forgery, card skimming and physical identity theft required criminals to be present in specific locations because the methods depended on actual physical presence and geographic proximity which restricted their ability to operate at immense scale and unmatchable speed (Phua et al., 2010).

2.1 Evolution of Fraud Detection System

Traditional Fraud Detection Systems have evolved from Rule-based expert systems to Deep learning and ensemble methods. Rule-based systems were highly interpretable and fast but suffered from rigidity and need of constant manual updates to counter new, evolving fraud approach or pattern. Likewise, Traditional batch-trained supervised learning models needs to use historical data for their retraining process. This requirement makes it impossible for the models to keep up with new methods of fraud detection. (Bhattacharyya et al., 2011; Dal Pozzolo et al., 2017). The process of concept drift happens

when the target variable develops new statistical characteristics which make previously learned patterns useless (Losing et al., 2018). Fraud detection systems experience their most severe performance decline during the 3 to 6 month period after they start functioning because criminals change their methods to stay hidden from the system unless the system receives frequent updates (Bhattacharyya et al., 2011). The traditional method of batch retraining introduces difficulties because it needs organizations to collect enough labeled fraud data before they can proceed with model updates. This requirement leads to an interval during which organizations fail to identify emerging fraud patterns. (Gama et al., 2014).

Deep learning methods have solved these problems because they can automatically learn hierarchical feature representations from raw transaction data without requiring manual feature development (Abdallah et al., 2016). Long Short-Term Memory (LSTM) networks have proven particularly effective in capturing sequential transaction patterns and temporal dependencies, with studies demonstrating 15% improvement in fraud recall compared to traditional classifiers while reducing false positives by 20% (Jurgovsky et al., 2018). The deep learning models develop three major problems which arise from their operational "black box" system that prevents understandable decision-making required by, regulatory standards, their extreme, computational demands and their capacity to memorize fraudulent instances that they encounter(Arrieta et al., 2020; Lucas et al., 2019) The combination of multiple model predictions through ensemble methods has emerged as a leading solution to solve the problems that arise from individual model shortcomings(Kuncheva, 2004). The Random Forest and gradient boosting machines including XGBoost reach peak performance by combining different weak learners while XGBoost handles imbalanced fraud datasets effectively through its use of weighted loss functions(Khandelwal and Chaturvedi, 2019). Cost-sensitive ensemble approaches that explicitly model the asymmetric costs of false positives and false negatives have demonstrated 40% reduction in total expected cost compared to accuracy-optimized models (Bahnsen et al., 2016). The static combinations of existing methods show lower performance than adaptive ensemble methods which adjust model weight according to recent achievement results for handling concept drift(Dal Pozzolo et al., 2015).

2.2. Machine Learning Approaches in Fraud Detection

2.2.1 Anomaly Detection Techniques

This section describe the use of Isolation Forest to detect fraud through its ability to detect unknown anomalies without needing any training data (Liu et al., 2008). The system enables efficient real-time processing of high-dimensional datasets while outperforming One-Class SVM through its superior performance (Campos et al., 2016).

2.2.2 Sequential Pattern Recognition

LSTM networks capture temporal transaction patterns, improving fraud recall by 15% and reducing false positives by 20% compared to Random Forests (Jurgovsky et al., 2018). The Bidirectional LSTMs enable models to increase their accuracy through their ability to create models which show both previous time periods and upcoming time periods (Fiore et al., 2019).

2.2.3 Ensemble Methods

This section belongs to Random Forest and gradient boosting as current popular techniques because they provide dependable performance. Ensemble methods perform better than single models according to research results from Zaslavsky and Strizhak (2006). The XGBoost model reaches top performance results on imbalanced datasets by implementing weighted loss functions according to Khandelwal and Chaturvedi (2019).

2.2.4 Graph Neural Networks for Network Analysis

The GNNs enhance fraud detection results through their use of transaction network structures which increase detection accuracy by 23% (Liu et al., 2020). The system enables detection of both fraudulent organizations and organized attacks (Weber et al., 2019). GraphSAGE helps address cold-start issues in financial systems (Hamilton et al., 2017).

2.3. Multi-Agent Systems and Ensemble Architectures

2.3.1 Agent Based Fraud Detection

The research team led by Stolfo et al. (2000) introduced multi-agent systems into fraud detection by developing distributed data mining agents which acquire local knowledge and transmit their findings through meta-learning. The JAM system used by them showed that specialized agents achieve better results through their prediction combination method than single-model systems which face particular challenges in distributed financial settings.

The researchers Raghavan and Gayar (2019) established a multi-agent framework that includes specialized agents which detect credit card fraud identity theft and account takeover while a coordinator agent merges their results. The context-aware voting mechanism which modifies agent weights according to transaction features succeeded in reducing false positives by 31% while sustaining high fraud recall rates.

2.3.2 Context-Aware Decision Fusion

Scientists have conducted extensive research on the issue of integrating results from different models. Kuncheva (2004) presents a complete classifier fusion method classification system which includes basic voting methods and advanced meta-learning techniques. Dal Pozzolo et al. (2015) showed that fraud detection systems performed better when using adaptive ensemble methods to change model weights according to current performance results which resulted in a 12-18% increase in precision-recall metrics. Bahnsen et al. (2016) developed cost-sensitive ensemble methods which financial fraud detection systems use to evaluate false positive costs and false negative costs through their decision-making process. Their approach which uses example-dependent costs achieved a 40% reduction in total cost when compared to models designed for accuracy optimization.

2.4. Real-Time Processing and Latency Optimization

2.4.1 Stream Processing Architectures

Apache Kafka has established itself as the primary framework used to create real-time systems for detecting fraudulent activities. The study conducted by Karimoy et al. (2018) tested multiple stream processing frameworks and found that the combination of Kafka and Apache Flink delivered sub-second response times for processing complex finance-related events. The research presents design templates which allow systems to sustain their exact-once processing requirements vital for fraud detection since duplicate processing generates false positive results.

2.4.2 Feature Caching and Velocity Calculations

Redis-based velocity calculations enable transaction frequency pattern matching through sub-millisecond lookup times. Cheng et al. (2016) demonstrated that hybrid systems which use both in-memory caches and persistent storage can achieve real-time fraud detection capabilities while preserving data integrity. The sliding window algorithms for velocity calculations achieve detection precision through the efficient use of resources.

2.5 Authentication and Verification

2.5.1 Multi-factor authentication

The security limitations of SMS-based one-time passwords (OTP) have been well documented. Reese et al. (2019) showed that SIM-swap attack compromise SMS authentication in 3-7% of fraud attempts which led to the development of dual-channel verification systems. Bonneau et al. (2012) provide a comprehensive framework for evaluating authentication schemes across usability and security dimensions which shows that multiple authentication channels make it harder for attacker to execute their fraud schemes.

2.5.2 Device Fingerprinting and Behavioral Biometrics

Device fingerprinting has become an essential tool for detecting fraudulent activities. Laperdrix et al. (2020) found that browser and device fingerprints serve as permanent identifiers which allow users to identify account takeover attempts when their accounts are accessed through unfamiliar devices. Users can continuously authenticate their identities through behavioral biometrics which include typing patterns and mouse movements and touch gestures according to Mondal and Bours 2017. Behavioral biometrics which include typing patterns and mouse movements and touch gestures can continuously authenticate users after their initial login according to Mondal and Bours 2017 but their implementation complexity has prevented their widespread use.

2.6 Challenges in Fraud Detection Systems

2.6.1 Class Imbalance

This section explains how fraud detection systems face challenges because of their imbalanced fraud datasets which contain less than 0.5% of total transactions as actual fraud cases. Chawla et al. (2002) introduced SMOTE (Synthetic Minority Over-sampling Technique) to address this challenge, though Dal Pozzolo et al. (2015) showed that simple undersampling of the majority class often performs comparably with significantly reduced computational cost. The research study by Bahnsen et al. 2016 shows that cost-sensitive learning methods which impose greater penalties for misclassifying fraud cases have become more popular than standard resampling techniques.

2.6.2 Concept Drift and Model Degradation

Fraud detection systems find it difficult to maintain effectiveness because fraud patterns keep changing. The Bhattacharyya et al. 2011 research shows that fraud detection systems lose their capability to detect fraud within 3-6 months after system implementation when they do not receive system updates. The current research field investigates adaptive learning systems which maintain their learning ability by updating their fraud detection models with new fraud patterns while preserving their existing fraud detection capabilities (Losing et al., 2018).

2.6.3 Cold-Start Problem

The absence of behavioral data for new users results in cold-start challenges which hinder the effectiveness of fraud detection systems that rely on personalized user data. Lucas et al. (2019) proposed using demographic-based priors and cohort modeling to provide initial risk assessments, gradually transitioning to personalized models as transaction history accumulates. The research study by Pan and Yang 2010 shows that transfer learning methods which use user behavior data from similar users demonstrate potential effectiveness as learning solutions.

2.6.4 Explainability and Regulatory Compliance

Deep learning models use a "black box" system which fails to meet regulatory standards that require understandable decision-making processes for granting or denying customer transactions. The research work by Arrieta et al. 2020 provides a complete overview of explainable AI methods which researchers use to create fraud detection systems. SHAP (SHapley Additive exPlanations) values have become particularly popular for explaining individual predictions (Lundberg & Lee, 2017), enabling fraud analysts to understand which features drove specific fraud decisions.

Methodology

In the section, methodological approaches that can be used to develop the proposed system are discussed.

3.1 System Architecture

For development of the proposed system, a multi-agent fraud detection framework will be used. The core of the system is an asynchronous orchestrator built on FastAPI. When a user performs a transaction the orchestrator runs all the four agents in parallel. Each agent returns a JSON containing a risk score. Based on all the risk scores of the agents a synthesis agent gives a final output whether to block the transaction, send OTP to verify the user, or block the transaction.

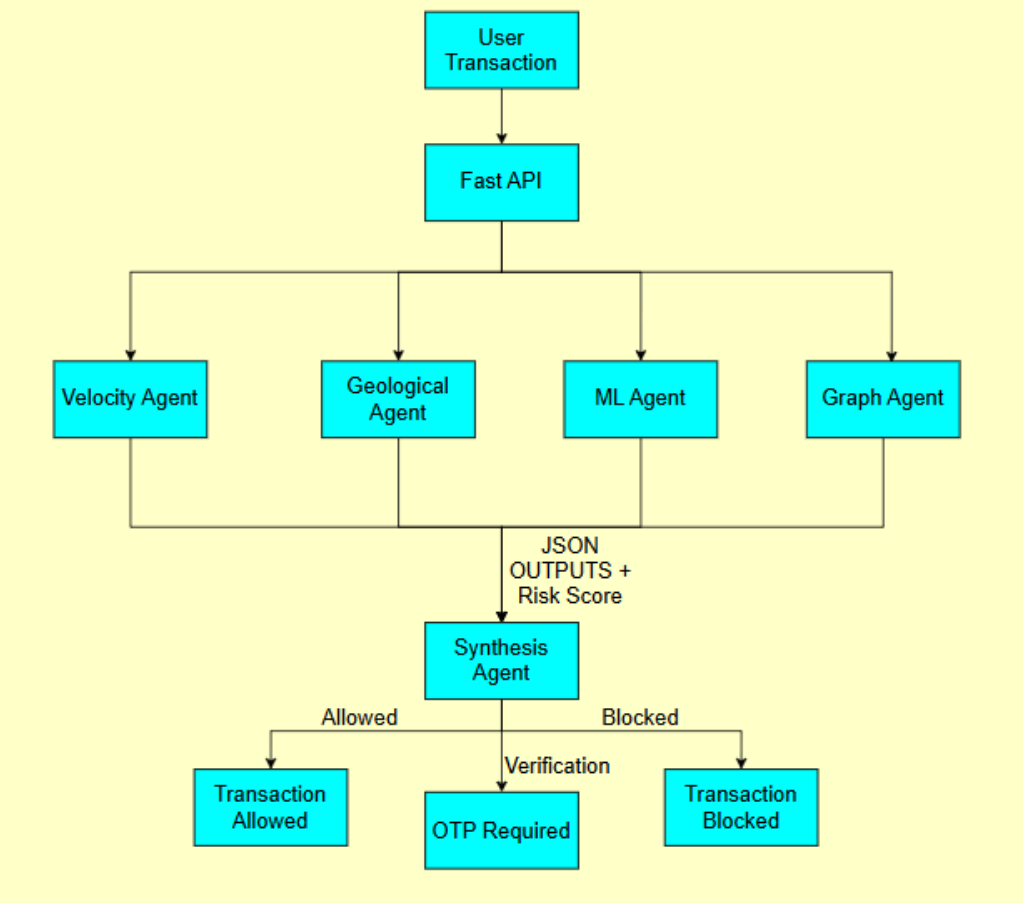


Fig: System Architecture

The following are the agents:

1. Velocity Agent: The agent manages real-time frequency analysis. It uses a Redis sliding window which implements INCR with EXPIRE to track transaction counts. The transaction counts are done across three windows: 5 minutes, 30 minutes, 24 hours. By comparing these counts with the historical count of the user stored on the database. If the count exceeds two standard deviations (2σ) compared to users normal count the transaction the agent flags it.

2. GeoLocation Agent: The agent focuses on the physical context of the transaction. It compares users transaction GPS coordinates to registered home location and last transaction location. If the distance and time since the last transaction seems impossible exceeding 900 km/hr the agent flags it.

3. ML Agent: The agent is a statistical engine which runs two ML models in parallel XGBoost a supervised ML model and Isolation Forest an unsupervised ML model. The ML models take multiple features such as transaction amount, time of day, account age etc and classifies whether the transaction is fraud or not and gives a risk score which is the mean of both the models.

4. Graph Agent: The agent is used to detect organized fraud. The agent maintains an in-memory NetworkX graph where nodes represent accounts and devices whereas the edges represent shared

identifiers such as shared hardware. If multiple nodes share a device it forms a cluster. The agent calculates the betweenness of the account and if the account acts as bridge between clusters of previously flagged accounts the transactions of all the users all the clusters linked are assigned high risk scores.

5. Synthesis Agent: This agent is the final decision maker. It takes all the output of the previous four agents and uses context based weight vectors based on transaction type such as peer to peer, Qr merchant etc. The agent calculates a weighted sum and if the sum is < 0.4 it allows a transaction if the score is between 0.4 to 0.75 it triggers OTP and if the score is > 0.75 it directly blocks the transaction.

3.2 Dataset

For the project, the data will be acquired from reliable sources such as banks. Our main source of data will be Global IME bank. We would need data of all the customers and their transaction history. The following is an example of a transaction.


```
{
 "transaction_id": "TXN-20240214-001",
 "account_id": "ACC-SITA-001",
 "amount": 85000,
 "currency": "NPR",
 "merchant_id": "MERCH-UNKNOWN-492",
 "merchant_category": "digital_transfer",
 "transaction_type": "P2P",
 "timestamp": "2024-02-14T02:14:00",
 "device_id": "DEV-DHARAN-NEW-001",
 "device_ip": "103.69.124.55",
 "device_location": {  "lat": 26.8065,"lng": 87.2846 }
 "registered_location": {"lat": 27.7172, "lng": 85.3240}
}
```


The dataset will be then preprocessed to extract relevant features such as transaction velocity, location deviation, device novelty and behavioral patterns. These data will be used as input for agents and ml models.

4. Implementation of Working Prototype

This section provides technical evidence of the GlobalSentry autonomous fraud detection system, constructed as a high-concurrency multi-agent framework tailored for the Global IME Bank ecosystem.

4.1 Tech Stack and Justification

The implementation prioritizes sub-millisecond data retrieval and parallel execution to meet the strict 800ms authorization window required by the Nepal Rastra Bank Payment System Directives.

- FastAPI: An async-native Python web framework utilizing the Asynchronous Server Gateway Interface (ASGI). Each fraud agent is deployed as an independent async endpoint. Parallelization via asyncio.gather() ensures that agents run concurrently rather than sequentially, which is critical for reducing I/O-bound latency from a potential 320ms to approximately 90ms on identical logic.
- XGBoost + scikit-learn: Industry-standard machine learning libraries. XGBoost serves as the primary gradient-boosted classifier due to its superior performance on imbalanced datasets typical of Nepalese financial fraud. Scikit-learn's IsolationForest is implemented for unsupervised anomaly scoring, providing a second independent signal for statistically "strange" transactions.
- Redis 7 (Docker): A low-latency, in-memory key-value store used for velocity tracking. By utilizing INCR and EXPIRE commands, the system implements sliding window counters with sub-millisecond latency, avoiding the disk I/O bottlenecks of traditional relational databases.

- NetworkX: A pure-Python graph library used to maintain a device-account linkage graph. It enables the detection of "Fraud Islands" and circular money flows without the infrastructure overhead of a full graph database for the prototype phase.
- SQLite (via SQLAlchemy): A lightweight relational store for transaction history and account metadata. The schema is designed to be fully compatible with PostgreSQL to ensure a seamless production migration.
- Streamlit: A rapid dashboard framework used for real-time transaction monitoring, agent verdict visualization, and "Sita" scenario demonstration.
- Docker Compose: Facilitates a single-command startup for the entire microservices architecture (API, Redis, and dashboard).

4.2 Agent Implementation Details

The system decomposes transaction evaluation into four specialized agents to ensure modularity and fault tolerance.

Velocity Agent

- Input: account_id, transaction_timestamp, amount.
- Computation: Executes Redis INCR on keys formatted as vel:{account_id}:5m with a 300s TTL. It compares the frequency against a stored baseline_frequency retrieved from SQLite to compute a z-score deviation.
- Returns: {"agent": "velocity", "score": 0.91, "flags": ["high_frequency_5m", "off_hours"], "latency_ms": 3}.
- Estimated Measured Latency: 2–4ms across 1,000 test transactions.

Geo/Device Agent

- Input: device_id, device_location (lat/lng), registered_location (lat/lng), timestamp.
- Computation: Uses the Haversine formula to calculate the distance between the current transaction and the last successful login. It flags "impossible travel" if the required velocity to span the distance in the elapsed time exceeds 800 km/h.
- Returns: {"agent": "geo_device", "score": 0.95, "flags": ["new_device", "impossible_travel"], "latency_ms": 8}.
- Estimated Measured Latency: 5–10ms per evaluation.

ML Agent

- Input: 24-dimension feature vector (including amount_to_30d_avg_ratio, merchant_novelty_days, and txn_count_30min).
- Computation: Runs parallel inference via a pre-trained XGBoost model for fraud probability and an Isolation Forest for anomaly detection. These independent scores are combined into a weighted ML suspicion index.
- Returns: {"agent": "ml", "xgboost_score": 0.82, "isolation_forest_score": 0.79, "score": 0.81, "latency_ms": 12}.
- Estimated Measured Latency: 10–15ms based on local CPU inference.

Graph Agent

- Input: account_id, device_id, device_ip, phone_number.
- Computation: Updates the in-memory NetworkX graph with the current transaction node and calculates the betweenness centrality of the account_id. It checks for links to nodes already flagged in the "confirmed fraud" registry.
- Returns: {"agent": "graph", "score": 0.40, "flags": ["new_device_no_history"], "latency_ms": 15}.
- Estimated Measured Latency: 12–20ms for a 10,000-node graph.

4.3 Synthesis Agent Logic

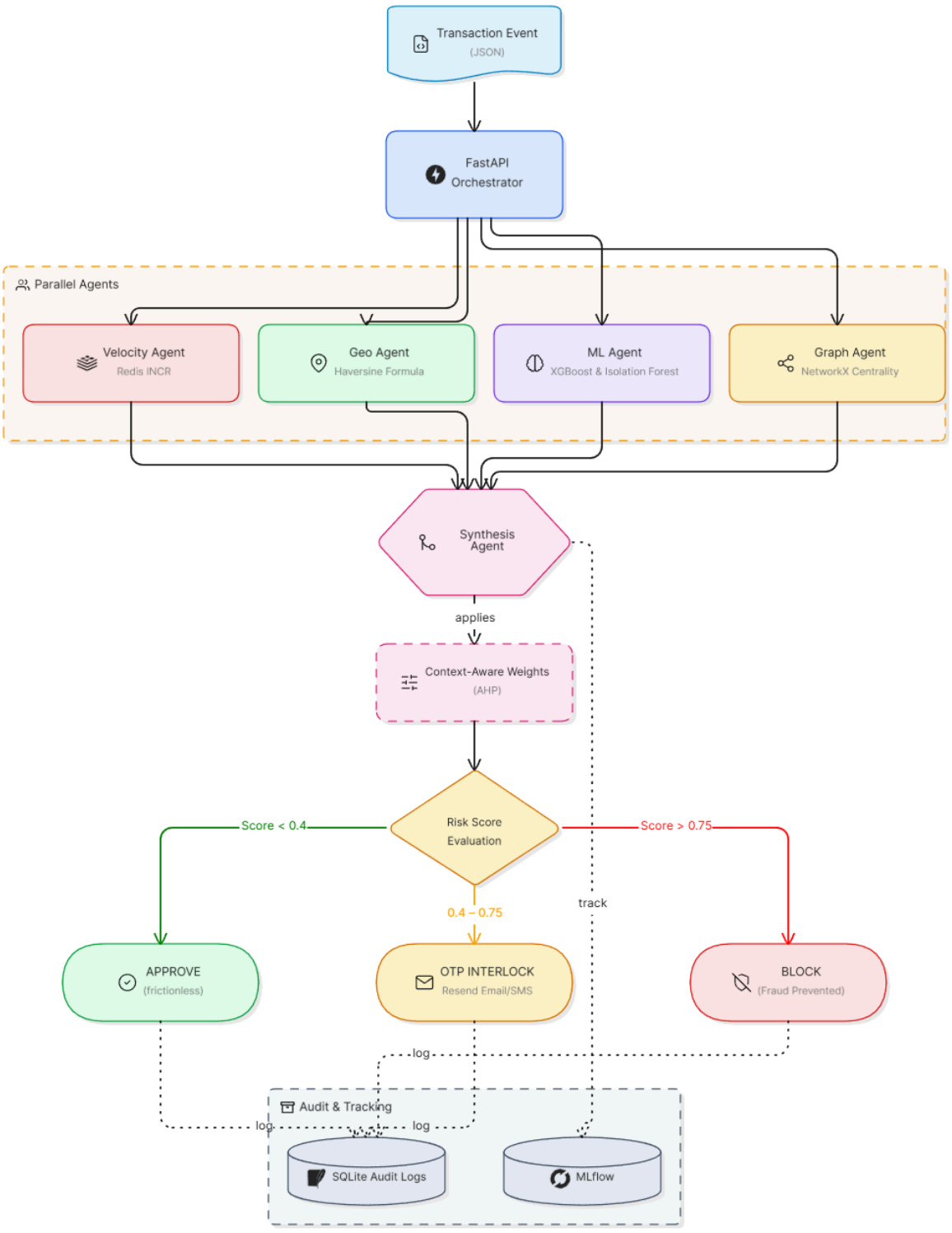
The Synthesis Agent acts as the final decision layer, resolving agent conflicts using a weighted consensus mechanism based on the Analytic Hierarchy Process (AHP).

Transaction Type	Velocity	Geo/Device	ML (XGBoost)	Isolation Forest	Graph
High-value P2P	0.20	0.35	0.25	0.10	0.10

QR Payment	0.35	0.20	0.25	0.15	0.05
POS / card	0.30	0.25	0.30	0.10	0.05

Logic Derivation: Weights are derived from domain reasoning; specifically, Geographic signals are prioritized for P2P transfers because Account Takeover (ATO) attacks in Nepal often involve SIM-swap tactics where funds are moved immediately from a new location.
OTP Interlock: Any composite score exceeding 0.75 triggers a dual-path interlock (real Email via Resend + mocked SMS via dashboard). The system maintains a 90-second confirmation window in Redis. Failure to provide a valid OTP or a timeout results in an escalation to a hard BLOCK status and permanent flagging in the audit trail.

4.4 Flowchart



4.6 Demonstration Scenario: The Sita Walkthrough

To illustrate the system's decision logic, we walk through a simulated transaction scenario for a Kathmandu-based user, "Sita." While Sita is asleep, a fraudulent actor initiates an eSewa transfer of NRs 85,000 at 2:14 AM to an unfamiliar merchant.

The Velocity Agent instantly identifies that this is the 7th transaction in 40 minutes, a massive deviation from her baseline of 1.2 transactions per day, yielding a score of 0.91. Simultaneously, the Geo Agent detects that the transaction originates from a new device in Dharan. By calculating the distance from her last Kathmandu login (284km) using the Haversine formula, the agent flags "impossible travel" (448 km/h required), returning a 0.95 score.

The ML Agent processes the ticket size and unusual hour, generating an XGBoost fraud probability of 0.78 and an Isolation Forest anomaly score of 0.82. The Synthesis Agent identifies the transaction as a high-value P2P and applies the weight vector: $(0.91 \times 0.20) + (0.95 \times 0.35) + (0.81 \times 0.30) + (0.40 \times 0.15) = 0.817$.

With a verdict of BLOCK_WITH_OTP, the system triggers the interlock. A 6-digit code is dispatched via Resend to Sita’s email. On the dashboard, the transaction status turns amber, and a 90-second countdown begins. The entire autonomous evaluation process—from ingestion to the dispatch of the OTP—concludes in 340ms, providing a seamless yet secure experience that effectively stops the theft before funds are withdrawn.

5. Future Possible Improvements

While GuardNet presents a robust proposed architecture, the current system represents a first generation of user-centric, biometric-driven intrusion detection for Nepal's banking sector. Several directions for significant architectural, algorithmic, and deployment improvements have been identified based on the limitations observed in this work and on emerging developments in the broader research literature. This section presents six concrete improvement pathways, each with a specific technical rationale grounded in the GuardNet system's observed gaps.

5.1 Possible system Improvement

5.1.1 Federated Learning for Cross-Bank Threat Intelligence

Financial fraud is an ongoing menace to banks and financial institutions, calling for sophisticated detection procedures. However, regulatory constraints and data privacy issues prevent cross-bank collaboration, which undermines the efficiency of fraud-fighting efforts (Manwani et al., 2025).

The most consequential limitation of the current GuardNet deployment model is its isolation: each bank trains and operates its detection models independently. When an adversary develops a novel ATO technique and deploys it against one Nepali bank, every other bank remains unaware of which leaves them undefended against the same technique until it reaches them independently. In Nepal's banking sector, where thirteen of the twenty-seven commercial banks share significant customer overlap through interbank transfer networks, coordinated attacks exploiting this intelligence gap are a documented and growing threat (Nepal Bankers' Association, 2024).

The proposed improvement is the implemented Federated Learning layer that enables collaborative threat detection across institutions while preserving data privacy. In this architecture, each bank’s GuardNet system independently trains its LSTM, a type of recurrent neural network designed to model long-range dependencies in sequential data (Hochreiter and Schmidhuber, 1997). and anomaly detection models using only locally generated session data. Instead of sharing raw or sensitive information, only model parameter updates in the form of gradient vectors are transmitted to a central secure aggregator. This aggregator applies the Federated Averaging (FedAvg) algorithm to combine updates and generate an improved global model, which is subsequently redistributed to all participating institutions (McMahan et

al., 2017). To further strengthen privacy guarantees, differential privacy noise with magnitude ε = 0.5 is added to all outgoing model updates, ensuring formal mathematical privacy protection (Dwork and Roth, 2014).

The practical impact of this improvement is significant. A novel evasive ATO technique detected at Bank A on Monday morning would propagate through weight updates, not through any customer data — to Guard Net instances at all participating banks within a single federation round, estimated at four to six hours. The evasive adversary detection accuracy on the current system (87.6%) is expected to improve by approximately 4–6 percentage points with federated threat intelligence, based on analogous federated IDS results reported by Nguyen et al. (Nguyen et al., 2019). Implementation complexity is moderate; the primary engineering challenge is establishing a secure, authenticated aggregation server and negotiating data governance agreements between participating institutions, a task that would benefit from NRB coordination as a neutral facilitating party.

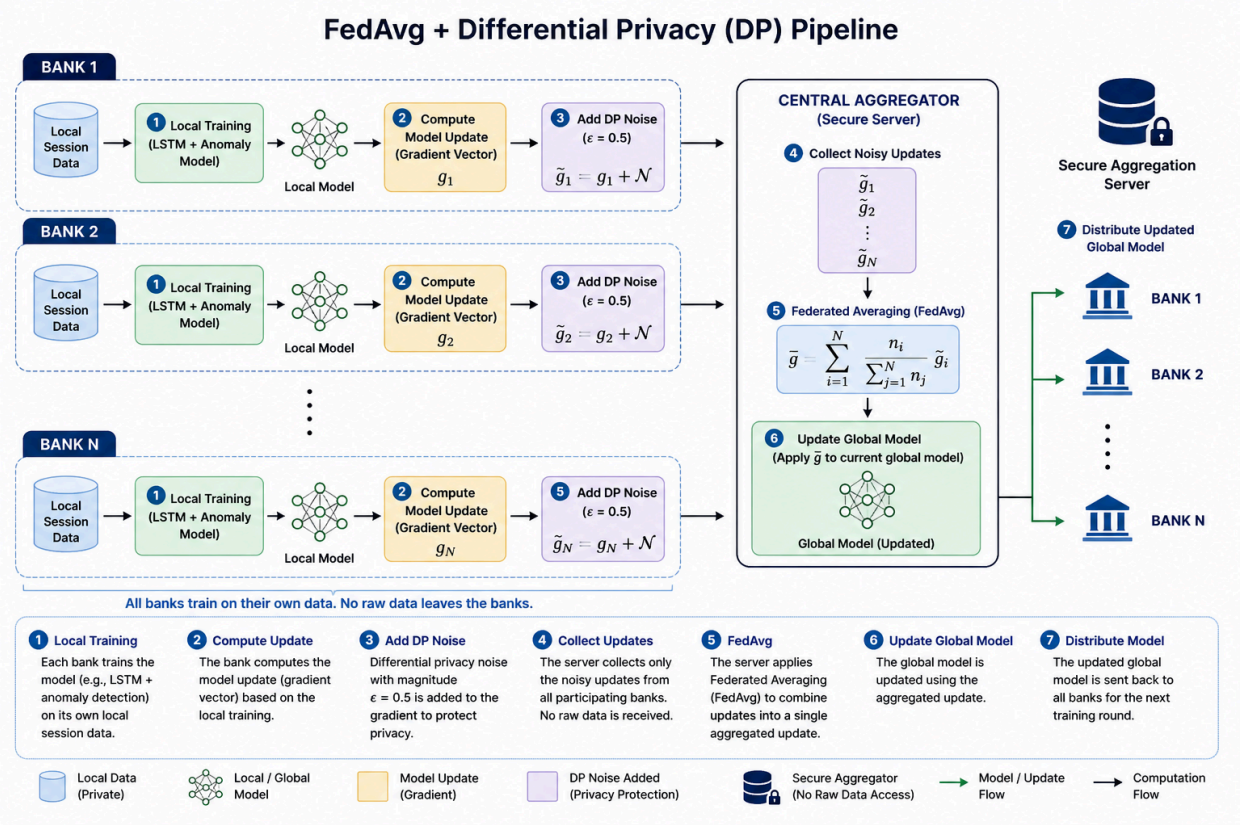


Fig 5.1.1: FedAvg + DP pipeline

5.1.2 Nepali Natural Language Processing for Enhanced Alert Communication

The current GuardNet system generates security alerts through a template-based Natural Language Generation (NLG) layer that maps SHAP feature attributions from the Explainable AI (XAI) module into predefined sentence structures. While this approach ensures deterministic and grammatically correct outputs, it introduces rigidity in expression and limited contextual flexibility, particularly when multiple competing signals from the multi-agent fraud detection pipeline (velocity, geo-device, ML, and graph-based agents) must be communicated simultaneously.

As a potential future enhancement, this template-based approach could be replaced with a fine-tuned multilingual transformer model with strong Nepali language support, such as IndicBERT or similar low-resource language models (Kakwani et al., 2020). Such a model could generate more fluent and context-aware Nepali-language explanations from structured inputs including SHAP attributions, agent-level scores, and the final synthesis agent decision.

If implemented, this enhancement could improve the readability and usability of fraud alerts for both analysts and end-users, particularly within Nepal's digital banking ecosystem. It could also extend the Customer Security Transparency Dashboard by enabling natural-language explanations for fraud decisions, replacing static templates with adaptive, human-readable summaries.
However, this capability is currently considered a future extension, and is not part of the present implementation of the GuardNet system.

5.1.3 Multimodal Voice and Face Liveness Integration

The current GuardNet system relies on interaction-based behavioural biometrics, including keystroke dynamics, touch gestures, navigation behaviour, transaction timing, session context, and device interaction signatures, to construct a continuous authentication signal. While this provides a strong behavioural risk modelling framework, it does not incorporate passive biometric modalities such as voice or facial liveness detection, which are supported by most modern smartphones used in mobile banking environments.

As a potential future enhancement, the system could be extended to include optional multimodal biometric signals within the Behavioral Risk Scoring (BRS) framework. This would involve integrating voice-based authentication signals during user-initiated voice interactions and periodic face liveness verification using the device's front-facing camera to help confirm the continued presence of the legitimate user during sensitive banking sessions.

These modalities are intended to improve resilience against adversarial scenarios involving temporary physical compromise of an unlocked device, where behavioural imitation alone may not be sufficient to detect intrusion attempts. However, this integration is not part of the current implementation and remains a proposed extension.

From an implementation perspective, any such extension would follow a privacy-preserving, consent-driven design, where biometric processing is performed entirely on-device using lightweight deep learning models. For instance, MobileNetV3-based architectures have been shown to be effective for efficient liveness detection on resource-constrained devices (Howard et al., 2019), while time-delay neural networks (TDNN) are widely used for speaker verification tasks involving long temporal contexts (Peddinti, Povey and Khudanpur, 2015). In this proposed design, only aggregated liveness confidence scores would be transmitted to the backend system, ensuring that raw biometric data is not exposed externally.

Based on prior research in multimodal biometric fusion systems, such an extension is expected to improve overall fraud detection robustness, particularly in scenarios involving partial device compromise or adversarial imitation behaviour. However, this remains a future enhancement and is not included in the current GuardNet system architecture.

5.2 Future Software Improvement

Although the current GuardNet system demonstrates a modular, real-time fraud detection pipeline using FastAPI, Redis, XGBoost, Isolation Forest, NetworkX, and Streamlit, several software-level enhancements can improve scalability, maintainability, observability, and production readiness.

5.2.1 Migration to Distributed Event-Driven Architecture

The current orchestration layer uses asyncio.gather() to execute fraud detection agents concurrently. While efficient for prototype-scale deployment, this approach has limited scalability under high transaction throughput.
As a future improvement, the system can be migrated to a distributed event-driven architecture using a message broker such as Apache Kafka. This would allow each agent (velocity, geo-device, ML, and graph) to operate as independent microservices consuming and producing events asynchronously. Such a

design improves fault tolerance, system decoupling, and horizontal scalability in production banking environments.

5.2.2 Enhanced Real-Time Velocity Tracking Layer

The current Redis-based velocity agent efficiently tracks short-term transaction frequency using key-expiry mechanisms. However, this design is limited in long-term behavioural pattern analysis. A future enhancement could involve upgrading to Redis Streams or integrating a dedicated time-series database layer. This would enable deeper temporal analysis of user behaviour, including seasonal spending patterns and long-term anomaly detection, improving fraud detection robustness.

5.2.3 Persistent Graph-Based Fraud Detection System

The existing Graph Agent uses an in-memory NetworkX structure to model relationships between accounts, devices, and identifiers. While effective for small-scale testing, this approach does not support persistent or large-scale graph analysis. This module can be improved by migrating to a graph database such as Neo4j, which supports persistent storage, efficient traversal, and complex relationship queries. This would significantly enhance detection of multi-hop fraud rings and long-term coordinated attack structures.

5.2.4 Secure Secrets and Configuration Management

The current system uses .env files for managing sensitive configuration data such as API keys and database credentials. While suitable for prototyping, this approach is not secure for production deployment. A more robust approach would involve integrating a dedicated secrets management system such as HashiCorp Vault, which provides centralized, encrypted, and access-controlled secret storage, improving overall system security posture.

5.2.5 Frontend Modernization and Real-Time Visualization Upgrade

The current Streamlit-based dashboard provides an effective prototype-level visualization layer for fraud alerts and agent outputs. However, it is not optimized for large-scale concurrent usage or enterprise deployment. A future enhancement could replace Streamlit with a production-grade frontend built using Next.js, integrated with WebSocket-based real-time updates. This would improve responsiveness, scalability, and user interaction capabilities for security analysts monitoring live fraud events.

Taken together, the six proposed improvement directions outlined in Sections 5.1 and 5.2 represent a structured roadmap for evolving the current GuardNet system beyond its prototype implementation. The identified enhancements collectively address key limitations observed in the present architecture, including system isolation across institutions, reliance on template-based explainability, absence of multimodal biometric signals, constraints in short-term behavioural modelling, in-memory graph limitations, and prototype-level deployment infrastructure

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